DATA ANALYSIS REPORT

Project / Case Study: Medical Appointment No-Show Analysis
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EXECUTIVE SUMMARY

Cedar Valley Community Health, a fictional outpatient healthcare organization, seeks to reduce missed patient appointments and improve scheduling efficiency. This analysis examined 110,527 historical medical appointment records to identify patterns associated with patient attendance, focusing on scheduling lead times, SMS reminders, patient demographics, and geographic characteristics.

The analysis identified an overall no-show rate of 20.19%, representing 22,319 missed appointments. Scheduling lead time showed one of the strongest observed relationships with attendance. Same-day appointments had a no-show rate of 4.65%, compared with 33.00% for appointments scheduled 31 or more days in advance. SMS reminders were associated with lower no-show rates when scheduling lead time was considered. Additionally, patients aged 18–29 had the highest no-show rate among the defined age groups at 24.65%, while neighborhood-level findings revealed differences in both no-show rates and missed-appointment volumes.

Based on these findings, the report recommends reducing unnecessary scheduling delays, piloting targeted SMS reminder strategies, and using patient demographic and geographic patterns to guide outreach efforts. These recommendations may help improve appointment attendance, scheduling efficiency, and patient access to care. Because the findings are based on historical observational data from Brazil, further validation using local appointment records is recommended before implementation.

1. BUSINESS TASK

Business Problem

Cedar Valley Community Health, a fictional outpatient healthcare organization, seeks to address missed patient appointments and improve scheduling efficiency. Patient no-shows can contribute to unused appointment slots, inefficient resource allocation, and reduced opportunities for other patients to receive timely care.

The organization needs to better understand appointment attendance patterns and identify factors associated with missed visits to support informed operational decisions.

Business Objective

Analyze historical medical appointment data to identify patterns and characteristics associated with patient no-shows. The analysis will investigate scheduling lead times, appointment weekdays, SMS reminders, and patient demographic and geographic characteristics.

The objective is to provide data-driven recommendations that may help clinic management improve appointment attendance, scheduling efficiency, and patient access to healthcare services.

Key Question(s)

- What is the overall appointment no-show rate, and how does attendance vary across appointment days and scheduling lead times?
- What relationship exists between SMS appointment reminders and patient attendance?
- Which patient demographic or geographic characteristics are associated with higher appointment no-show rates, and how can these findings inform improvements to clinic scheduling and patient access?

Stakeholders

Marcus Chen — Director of Clinic Operations
Oversees operational performance across Cedar Valley Community Health. Interested in understanding appointment no-show rates, improving scheduling efficiency, and identifying opportunities to optimize clinic resources.

Sophia Bennett — Patient Scheduling Manager
Responsible for appointment scheduling, reminders, cancellations, and rescheduling procedures. Seeks to identify scheduling patterns associated with missed appointments and evaluate potential improvements to appointment management.

Dr. Elena Martinez — Medical Director
Oversees clinical services and patient care quality. Interested in understanding attendance patterns that may affect continuity of care and patient access to medical services.

Aisha Patel — Patient Access Coordinator
Supports patient scheduling, communication, and appointment accessibility. Interested in identifying attendance disparities and potential opportunities to address barriers to appointment attendance.

Primary Audience: Director of Clinic Operations and Patient Scheduling Manager.
Secondary Audience: Medical Director and Patient Access Coordinator.

2. DATA SOURCES

Data Source 1 — Medical Appointment No-Shows

Source: Joni Hoppen (JoniHoppen), with attribution to Aquarela Analytics.
https://www.kaggle.com/datasets/joniarroba/noshowappointments
Format: CSV file, imported into PostgreSQL using DBeaver
Time Period: Appointment dates from April 29, 2016, to June 8, 2016. Scheduling records extend from November 10, 2015, to June 8, 2016.
Size: 110,527 appointment records and 14 columns
Description: The Medical Appointment No Shows dataset contains historical healthcare appointment records from Brazil. Each record represents a scheduled medical appointment and includes information about patient demographics, appointment scheduling dates, medical conditions, SMS reminders, and attendance outcomes. The dataset is useful for investigating patterns associated with missed appointments and identifying potential relationships between patient characteristics, scheduling practices, and attendance. For Cedar Valley Community Health, these records provide a foundation for exploring appointment no-show patterns and developing data-informed recommendations to improve patient attendance.

Data Reliability and Limitations

The Medical Appointment No Shows dataset was obtained from Kaggle and published by Joni Hoppen, with attribution to Aquarela Analytics. The dataset is publicly available under the Creative Commons Attribution-NonCommercial-ShareAlike 4.0 (CC BY-NC-SA 4.0) license. Although the dataset provides useful historical healthcare appointment information, its original data collection methods and completeness cannot be independently verified from the available documentation.

Initial data-integrity checks in PostgreSQL confirmed that the dataset contains 110,527 appointment records and 14 columns. No duplicate appointment IDs were identified, and no missing values were found in five essential fields: patient ID, appointment ID, scheduled date, appointment date, and no-show status. However, six records contained unusual or invalid ages, including one negative age and five records with an age of 115. Additionally, five appointments had appointment dates preceding their scheduling dates. These issues will be investigated further during the Process Phase.

Several limitations may affect the interpretation and generalizability of the findings. The dataset contains historical records from Brazil, which may not accurately represent current healthcare practices or patient populations in other regions. Approximately 65% of appointment records involve female patients, and individual patients may appear in multiple records. Additionally, the dataset does not include potentially important factors such as transportation access, appointment type, or patients' reasons for missing appointments. The handcap variable also contains values from 0 to 4 without sufficiently clear documentation of their meaning.

Despite these limitations, the dataset provides a useful foundation for investigating patterns associated with missed medical appointments. Findings will be interpreted as associations rather than evidence of causation, and the limitations will be considered when developing recommendations for Cedar Valley Community Health.

3. DATA CLEANING AND MANIPULATION

Tools Used

PostgreSQL and DBeaver were used throughout the Process Phase. PostgreSQL served as the relational database management system for storing, cleaning, transforming, and validating the appointment dataset using SQL. DBeaver was used as the SQL development environment to write and execute queries, inspect records, investigate data-quality issues, and document the cleaning procedures.

Cleaning Process

- **Removed duplicate records:** Examined all 110,527 appointment records for duplicate appointment IDs. No duplicates were identified, so no records were removed. All original appointments were preserved.
- **Handled missing or null values:** Checked all 14 original columns for NULL values, empty strings, and whitespace-only entries. No missing values were detected in the original dataset. One record containing an invalid patient age of -1 was converted to NULL after determining that the correct age could not be verified. Additionally, five appointments with inconsistent scheduling dates were assigned NULL values in the calculated lead-time column while preserving their original dates and appointment records.
- **Corrected data types and formatting:** Created a separate working table, `appointments_clean`, from the original `appointments_raw` table. Converted text-based fields into appropriate PostgreSQL data types, including BIGINT for appointment IDs, INTEGER for ages, SMALLINT for binary indicators, TIMESTAMPTZ for scheduling timestamps, and DATE for appointment dates. Patient IDs remained as TEXT to preserve their identifier values.
- **Standardized column names and categories:** Renamed the original `hipertension` column to `hypertension` for improved consistency and readability. Verified that gender, scholarship, hypertension, diabetes, alcoholism, SMS notification, and appointment outcome fields contained expected categories. Examined 81 unique neighborhood names and found no capitalization or whitespace inconsistencies requiring correction.
- **Investigated invalid values and outliers:** Identified and investigated one invalid age of -1 and five appointments recorded with an age of 115. The invalid negative age was replaced with NULL. The five age-115 appointments, belonging to two patients, were retained because their recorded ages were internally consistent and could not be

established as incorrect. Investigated the `handcap` column, which contained values ranging from 0 to 4, and preserved all values because their exact meanings could not be independently verified. Also identified five appointments with scheduling dates occurring after their appointment dates.

- **Created calculated fields and new variables:** Created `lead_days` to measure the number of calendar days between appointment scheduling and the scheduled appointment date. Used a CASE expression to calculate non-negative lead times and assign NULL to five invalid negative lead times. Created `no_show_flag`, converting the original appointment outcome categories into numeric indicators (1 = missed appointment; 0 = did not miss appointment) to support subsequent analysis.
- **Joined or merged datasets:** No dataset merging was necessary because the project used one primary appointment dataset. SQL JOIN operations were used during validation to compare the raw and cleaned tables and verify that appointment records were preserved correctly.

Quality Checks

Data integrity was verified through multiple SQL validation queries performed before and after cleaning.

The cleaned dataset retained all **110,527 appointment records**, with 110,527 unique appointment IDs and no duplicate IDs. A FULL OUTER JOIN comparison confirmed that every appointment ID in the original dataset was present in the cleaned table, with no missing or unexpected appointments.

Field-level comparisons between `appointments_raw` and `appointments_clean` identified zero unexpected differences across the original 14 columns after accounting for intentional transformations.

Additional checks confirmed that no negative patient ages remained, one invalid age had been converted to NULL, and five invalid scheduling lead times had been assigned NULL. Of the 110,527 appointments, 110,522 contained valid calculated lead times, with zero negative or mismatched lead-time values.

The numeric appointment outcome transformation was also validated. All 88,208 appointments marked “No” were correctly assigned a `no_show_flag` of 0, while all 22,319 appointments marked “Yes” were assigned a value of 1. No incorrect mappings were detected.

Categorical values and neighborhood formatting were reviewed, and the distribution of `handcap` values was confirmed to match the original dataset.

All cleaning operations were documented in the SQL script `02_process_cleaning.sql`, and the original `appointments_raw` table was preserved unchanged.

These checks confirmed that the documented transformations were applied consistently and that the cleaned dataset was ready for the Analyze Phase, while retaining noted limitations concerning extreme ages, uncertain disability coding, and invalid scheduling information.

4. ANALYSIS

Approach

PostgreSQL and DBeaver were used to conduct exploratory and descriptive analysis of 110,527 medical appointment records. SQL queries were developed to calculate appointment volumes, attendance rates, and no-show percentages across patient demographics, medical characteristics, scheduling lead times, SMS reminder status, neighborhoods, and appointment dates.

Analytical techniques included conditional aggregation, Common Table Expressions (CTEs), grouping, filtering, date functions, and percentage calculations. These methods were appropriate for identifying appointment attendance patterns, comparing patient groups, and investigating factors associated with missed appointments.

Additional analyses examined scheduling lead time alongside SMS reminder status to investigate potentially misleading relationships. Aggregated results were exported as CSV files for further visualization in Tableau.

Analysis Steps

1. **Establish Baseline Appointment Metrics:** Calculated total appointments, unique patients, appointment date range, attended appointments, missed appointments, and the overall no-show rate.
2. **Analyze Patient Demographics:** Compared appointment volumes and no-show rates across gender and age groups to identify demographic differences in appointment attendance.
3. **Investigate Scheduling Lead Times:** Categorized appointments by the number of days between scheduling and the appointment date. Calculated no-show rates for same-day, 1–7 day, 8–14 day, 15–30 day, and 31+ day scheduling intervals.
4. **Evaluate SMS Reminder Patterns:** Compared appointment attendance by SMS reminder status, examined SMS reminders within scheduling lead-time categories, and repeated the comparison after excluding same-day appointments.
5. **Examine Patient Assistance and Medical Characteristics:** Analyzed no-show rates by scholarship participation, hypertension, diabetes, alcoholism indicator, and recorded handicap categories.
6. **Investigate Geographic Differences:** Calculated neighborhood-level appointment volumes and no-show rates. Ranked neighborhoods by both their no-show percentages and absolute numbers of missed appointments, using minimum sample-size thresholds for exploratory comparisons.

7. **Analyze Appointment Weekdays:** Compared appointment volumes and no-show rates across days of the week and verified that no Sunday appointments were recorded.
8. **Evaluate Daily and Weekly Trends:** Aggregated appointment records by date and calendar week to examine fluctuations in appointment volumes and no-show rates throughout the observation period.
9. **Organize and Export Analysis Results:** Created aggregated SQL summaries and exported seven CSV files containing neighborhood, daily, weekly, scheduling lead-time, SMS reminder, age-group, and weekday results for further visualization in Tableau.

Results

The analysis identified an overall appointment no-show rate of **20.19%**, representing 22,319 missed appointments out of 110,527 scheduled appointments. Age differences were more pronounced than gender differences. Patients aged 18–29 had the highest no-show rate among the defined age groups at 24.65%, while patients aged 60–74 had the lowest at 15.04%. Gender-based no-show rates were relatively similar.

Scheduling lead time demonstrated one of the strongest observed relationships with appointment attendance. Same-day appointments had a no-show rate of 4.65%, compared with 33.00% for appointments scheduled more than 30 days in advance. This suggests that appointments booked farther in advance may warrant additional attention when developing strategies to reduce missed appointments.

SMS reminder analysis revealed an important difference between aggregated and grouped results. Although appointments associated with SMS reminders initially showed higher overall no-show rates, comparisons within scheduling lead-time categories showed lower rates among appointments marked as receiving reminders. After excluding same-day appointments, the SMS group had a no-show rate of 27.57%, compared with 29.44% for appointments without SMS reminders. These findings highlight the importance of considering scheduling lead time when evaluating reminder-related attendance patterns.

Patient characteristics showed additional differences. Scholarship participants had a higher no-show rate (23.74%) than nonparticipants (19.81%). Patients recorded as having hypertension or diabetes had lower observed no-show rates than patients without those conditions. The alcoholism indicator showed almost no difference, while several handicap categories contained too few appointments to support meaningful comparisons.

Geographic analysis revealed that neighborhoods with the highest no-show rates were not necessarily those with the largest numbers of missed appointments. Santos Dumont had a comparatively high no-show rate of 28.92%, while Jardim Camburi recorded the most missed appointments (1,465) despite having a below-average no-show rate of 18.98%. This distinction demonstrates the importance of considering both missed-appointment percentages and appointment volumes when evaluating operational priorities.

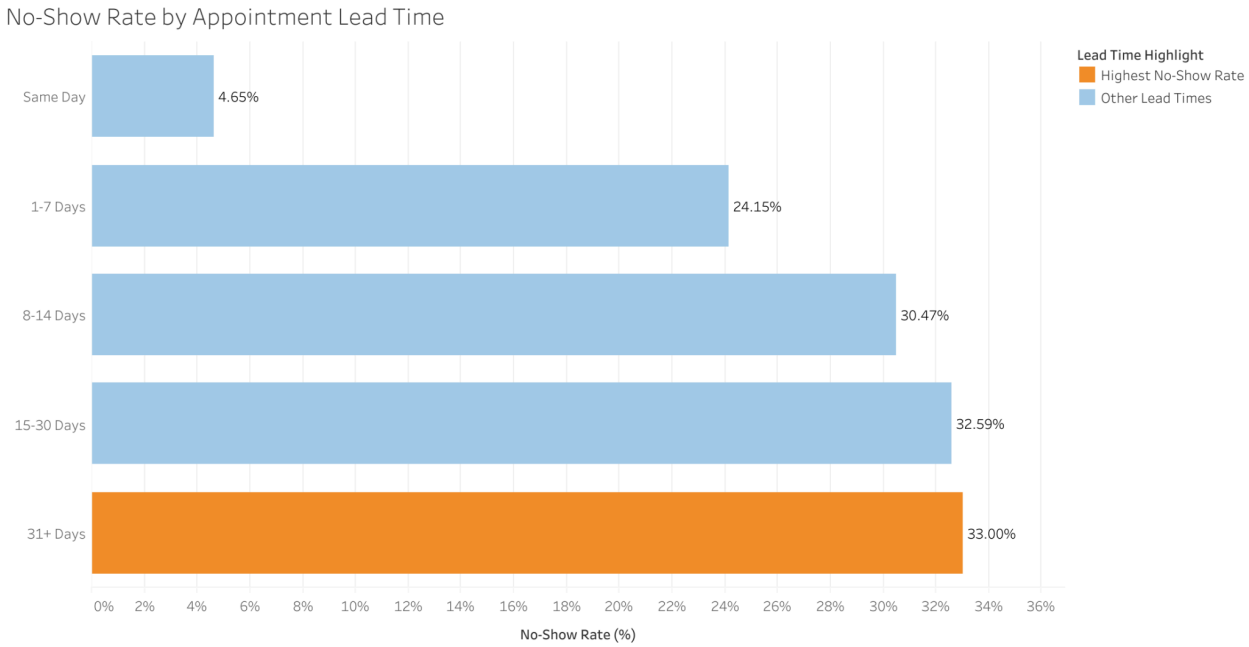
Appointment attendance also varied over time. Friday had the highest no-show rate among Monday–Friday appointments at 21.23%. Weekly no-show rates peaked at 21.74% during the

week beginning May 16, 2016, before declining to 18.13% during the final observed week beginning June 6. However, the short observation period and incomplete calendar weeks limit conclusions about long-term trends.

Overall, **scheduling lead time, age, SMS reminder status, and neighborhood-level differences emerged as important areas for further investigation**. These findings provide Cedar Valley Community Health with evidence to guide the development of potential appointment attendance strategies. However, the results describe associations rather than causal relationships, and additional investigation would be necessary before implementing targeted interventions.

5. SUPPORTING VISUALIZATIONS AND KEY FINDINGS

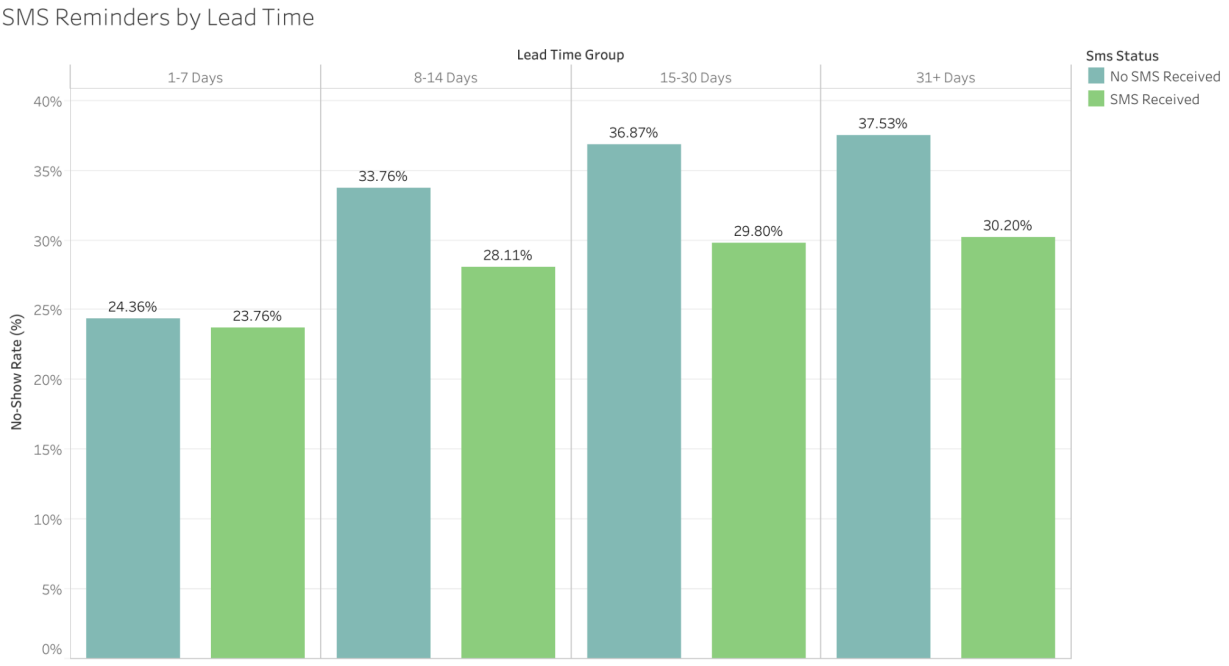
Figure 1. No-Show Rate by Appointment Lead Time



Key Finding

Scheduling lead time showed a strong association with appointment attendance. Same-day appointments had the lowest no-show rate at 4.65%, compared with 24.15% for appointments scheduled 1–7 days in advance, 30.47% for 8–14 days, 32.59% for 15–30 days, and 33.00% for 31 or more days. This pattern indicates that appointments scheduled farther in advance were associated with a greater likelihood of being missed. The findings suggest that reducing unnecessary scheduling delays and providing additional follow-up for appointments booked well in advance may be valuable strategies for improving attendance.

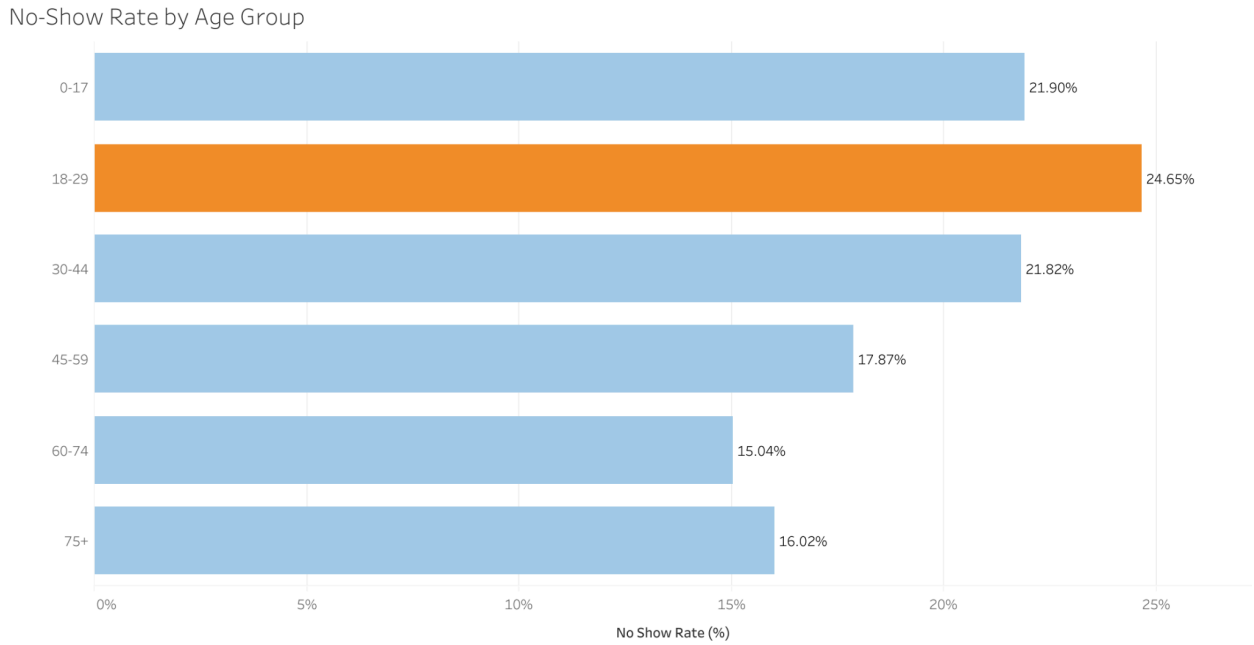
Figure 2. SMS Reminder Status by Appointment Lead Time



Key Finding

When appointments were compared within the same scheduling lead-time categories, patients recorded as receiving SMS reminders consistently had lower no-show rates than those without reminders. The difference was particularly noticeable for appointments scheduled 31 or more days in advance, where the no-show rate was 30.20% among appointments with SMS reminders compared with 37.53% among those without reminders. After excluding same-day appointments, the overall no-show rate was 27.57% for appointments with SMS reminders versus 29.44% for those without. These findings demonstrate the importance of accounting for scheduling lead time when evaluating reminder-related attendance patterns. Although the results do not establish that SMS reminders caused improved attendance, they suggest that reminder strategies warrant further evaluation, particularly for appointments scheduled farther in advance.

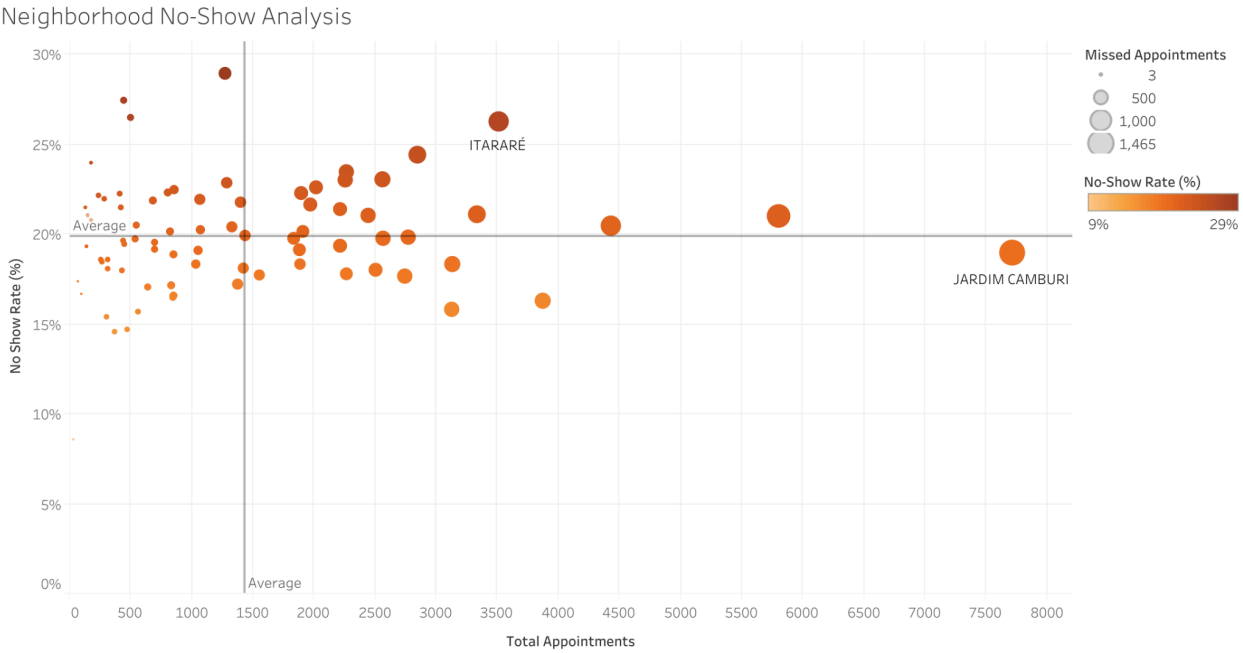
Figure 3. No-Show Rate by Age Group



Key Finding

Appointment no-show rates varied across patient age groups, with younger adults generally exhibiting higher rates than older adults. Patients aged 18–29 had the highest no-show rate at 24.65%, while patients aged 60–74 had the lowest at 15.04%, representing a difference of 9.61 percentage points. Patients aged 75 and older also had a comparatively low rate of 16.02%. These differences suggest that age is an important characteristic to consider when investigating appointment attendance patterns. Further research into potential scheduling, communication, and accessibility barriers among younger adults may help identify opportunities for more effective patient support.

Figure 4. Neighborhood No-Show Rates and Appointment Volumes



Key Finding

Neighborhood-level analysis revealed that areas with higher no-show rates did not necessarily generate the largest numbers of missed appointments. For example, Santos Dumont had a relatively high no-show rate of 28.92%, while Jardim Camburi recorded the highest absolute number of missed appointments (1,465) despite having a below-average no-show rate of 18.98%. This distinction highlights the importance of evaluating both appointment volume and no-show percentages when identifying operational priorities. Neighborhoods with elevated no-show rates may warrant further investigation into potential attendance barriers, while high-volume neighborhoods may present opportunities for substantial reductions in missed appointments even when their rates are comparatively lower.

Overall Key Findings

- **Scheduling lead time was a major factor associated with appointment attendance.** Across 110,527 appointments, 22,319 were missed, resulting in an overall no-show rate of 20.19%. No-show rates increased substantially as scheduling lead times lengthened, suggesting that appointment timing and follow-up procedures are important areas for operational improvement.
- **SMS reminders were associated with lower no-show rates when scheduling lead time was considered.** Comparisons within lead-time categories revealed consistently lower no-show rates among appointments with SMS reminders. This suggests that reminder strategies, particularly for appointments scheduled farther in advance, deserve further investigation and evaluation.

- **Patient demographics and geographic characteristics revealed additional differences in appointment attendance.** Young adults aged 18–29 had the highest no-show rate among the defined age groups, while neighborhood comparisons demonstrated that high no-show percentages and large numbers of missed appointments are not necessarily concentrated in the same areas. These findings support further investigation of patient attendance barriers and the development of targeted, evidence-informed operational strategies.

6. TOP THREE RECOMMENDATIONS

Recommendation 1 — Reduce Scheduling Delays and Reconfirm Long-Wait Appointments

Recommendation: Review appointment capacity and offer earlier openings, including cancellations, whenever clinically appropriate. Introduce a structured confirmation and rescheduling check-in for appointments booked 15 or more days ahead, with additional attention to those scheduled 31 or more days in advance.

Evidence: No-show rates rose from 4.65% for same-day appointments to 32.59% for appointments scheduled 15–30 days ahead and 33.00% for those scheduled 31 or more days ahead. This was one of the clearest patterns identified in the analysis.

Expected Benefit: Shorter waits and earlier confirmation may reduce missed visits, help staff reassign appointment slots that patients cannot use, and improve scheduling efficiency and patient access.

Recommendation 2 — Pilot Lead-Time-Based SMS Reminders

Recommendation: Test an opt-in SMS reminder process that gives patients a simple way to confirm or reschedule upcoming visits, prioritizing appointments booked well in advance. Compare outcomes among patients with similar scheduling lead times and evaluate the pilot before expanding it.

Evidence: For appointments scheduled 31 or more days ahead, the no-show rate was 30.20% among those with SMS reminders compared with 37.53% among those without. After excluding same-day appointments, the rates were 27.57% and 29.44%, respectively.

Expected Benefit: More timely communication may encourage confirmations and earlier cancellations, allowing staff to refill available slots. Evaluating the pilot would help determine whether the reminder process improves attendance.

Recommendation 3 — Target Outreach Using Local Attendance Patterns

Recommendation: Validate age-group and neighborhood patterns using Cedar Valley Community Health's own recent appointment data. Then pilot tailored outreach for groups with elevated no-show rates or high numbers of missed visits, gathering patient feedback to identify scheduling, communication, and access barriers.

Evidence: Patients aged 18–29 had a 24.65% no-show rate, compared with 15.04% for ages 60–74. Santos Dumont had a 28.92% no-show rate, while Jardim Camburi had 1,465 missed appointments despite a lower rate of 18.98%. These results show why both percentages and counts matter.

Expected Benefit: Locally validated outreach could help the clinic focus resources where they are most needed, address barriers to attendance, and support more equitable patient access without relying on a one-size-fits-all approach.

CONCLUSION

This analysis examined 110,527 historical medical appointments to identify patterns associated with missed visits and inform scheduling improvements for Cedar Valley Community Health. Overall, 22,319 appointments were missed (20.19%). Longer scheduling lead times were associated with substantially higher no-show rates; SMS reminders were associated with lower rates after accounting for lead time; and attendance differed by age group and neighborhood. The recommendations address these findings by reducing avoidable scheduling delays, testing targeted SMS reminders, and using local attendance data to guide outreach. Because the dataset describes appointments in Brazil in 2016 and the analysis is observational, Cedar Valley should validate these patterns locally and assess pilot outcomes before broader implementation.

APPENDIX

APPENDIX A — SUPPLEMENTARY DATA TABLES

Table A.1 — Appointment Counts by Scheduling Lead Time

lead_time_group	total_appointments	attended_appointments	missed_appointments	no_show_rate_pct	attended_rate_pct
Same Day	38,563	36,771	1,792	4.65%	95.35%
1-7 Days	32,185	24,413	7,772	24.15%	75.85%
8-14 Days	12,025	8,361	3,664	30.47%	69.53%
15-30 Days	17,371	11,710	5,661	32.59%	67.41%
31+ Days	10,378	6,953	3,425	33%	67%
Unknown	5	0	5	100%	0%

Table A.2 — Appointment Counts by SMS Status and Lead Time

lead_time_group	sms_status	total_appointments	attended_appointments	missed_appointments	no_show_rate_pct	attended_rate_pct
Same Day	No SMS Received	38,563	36,771	1,792	4.65%	95.35%
1-7 Days	SMS Received	11,548	8,804	2,744	23.76%	76.24%
1-7 Days	No SMS Received	20,637	15,609	5,028	24.36%	75.64%
8-14 Days	SMS Received	7,004	5,035	1,969	28.11%	71.89%
8-14 Days	No SMS Received	5,021	3,326	1,695	33.76%	66.24%
15-30 Days	SMS Received	10,520	7,385	3,135	29.80%	70.20%
15-30 Days	No SMS Received	6,851	4,325	2,526	36.87%	63.13%
31+ Days	SMS Received	6,410	4,474	1,936	30.20%	69.80%
31+ Days	No SMS Received	3,968	2,479	1,489	37.53%	62.47%
Unknown	No SMS Received	5	0	5	100%	0%

Table A.3 — Appointment Counts by Age Group

age_group	total_appointments	attended_appointments	missed_appointments	no_show_rate_pct
0-17	27,379	21,382	5,997	21.90%
18-29	16,731	12,607	4,124	24.65%
30-44	22,022	17,216	4,806	21.82%
45-59	23,221	19,071	4,150	17.87%
60-74	15,237	12,946	2,291	15.04%
75+	5,936	4,985	951	16.02%
Unknown	1	1	0	0%

Table A.4 — Complete Neighborhood Appointment Data

neighbourhood	total_appointments	attended_appointments	missed_appointments	no_show_rate_pct	attended_rate_pct
JARDIM CAMBURI	7,717	6,252	1,465	18.98%	81.02%
MARIA ORTIZ	5,805	4,586	1,219	21%	79%
RESISTÊNCIA	4,431	3,525	906	20.45%	79.55%
JARDIM DA PENHA	3,877	3,246	631	16.28%	83.72%
ITARARÉ	3,514	2,591	923	26.27%	73.73%
CENTRO	3,334	2,631	703	21.09%	78.91%
TABUAZEIRO	3,132	2,559	573	18.30%	81.7%
SANTA MARTHA	3,131	2,635	496	15.84%	84.16%
JESUS DE NAZARETH	2,853	2,157	696	24.40%	75.60%
BONFIM	2,773	2,223	550	19.83%	80.17%
SANTO ANTÔNIO	2,746	2,262	484	17.63%	82.37%
SANTO ANDRÉ	2,571	2,063	508	19.76%	80.24%
CARATOÍRA	2,565	1,974	591	23.04%	76.96%
JABOUR	2,509	2,058	451	17.98%	82.02%
SÃO PEDRO	2,448	1,933	515	21.04%	78.96%
ILHA DO PRÍNCIPE	2,266	1,734	532	23.48%	76.52%
NOVA PALESTINA	2,264	1,862	402	17.76%	82.24%
ANDORINHAS	2,262	1,741	521	23.03%	76.97%
DA PENHA	2,217	1,788	429	19.35%	80.65%
ROMÃO	2,215	1,741	474	21.40%	78.60%
GURIGICA	2,018	1,562	456	22.60%	77.40%

SÃO JOSÉ	1,977	1,549	428	21.65%	78.35%
BELA VISTA	1,907	1,523	384	20.14%	79.86%
MARUÍPE	1,902	1,478	424	22.29%	77.71%
FORTE SÃO JOÃO	1,889	1,543	346	18.32%	81.68%
ILHA DE SANTA MARIA	1,885	1,524	361	19.15%	80.85%
SÃO CRISTÓVÃO	1,836	1,473	363	19.77%	80.23%
REDENÇÃO	1,553	1,278	275	17.71%	82.29%
SÃO BENEDITO	1,439	1,152	287	19.94%	80.06%
JOANA D´ARC	1,427	1,169	258	18.08%	81.92%
CRUZAMENTO	1,398	1,094	304	21.75%	78.25%
CONSOLAÇÃO	1,376	1,139	237	17.22%	82.78%
SANTA TEREZA	1,332	1,060	272	20.42%	79.58%
PRAIA DO SUÁ	1,288	994	294	22.83%	77.17%
SANTOS DUMONT	1,276	907	369	28.92%	71.08%
ILHA DAS CAIEIRAS	1,071	836	235	21.94%	78.06%
GRANDE VITÓRIA	1,071	854	217	20.26%	79.74%
INHANGUETÁ	1,057	855	202	19.11%	80.89%
PRAIA DO CANTO	1,035	845	190	18.36%	81.64%
BENTO FERREIRA	858	665	193	22.49%	77.51%
VILA RUBIM	851	710	141	16.57%	83.43%
DO QUADRO	849	709	140	16.49%	83.51%
CONQUISTA	849	689	160	18.85%	81.15%
REPÚBLICA	835	692	143	17.13%	82.87%
MONTE BELO	824	658	166	20.15%	79.85%
PARQUE MOSCOSO	802	623	179	22.32%	77.68%
GOIABEIRAS	700	563	137	19.57%	80.43%
JUCUTUQUARA	694	561	133	19.16%	80.84%
FONTE GRANDE	682	533	149	21.85%	78.15%
MATA DA PRAIA	644	534	110	17.08%	82.92%
DO CABRAL	560	472	88	15.71%	84.29%
SANTOS REIS	547	435	112	20.48%	79.52%
ESTRELINHA	538	432	106	19.70%	80.30%
SANTA CLARA	506	372	134	26.48%	73.52%

SOLON BORGES	469	400	69	14.71%	85.29%
PIEDADE	452	364	88	19.47%	80.53%
SANTA CECÍLIA	448	325	123	27.46%	72.54%
SANTA LÚCIA	438	352	86	19.63%	80.37%
SANTA LUÍZA	428	351	77	17.99%	82.01%
BARRO VERMELHO	423	332	91	21.51%	78.49%
DO MOSCOSO	413	321	92	22.28%	77.72%
MÁRIO CYPRESTE	371	317	54	14.56%	85.44%
BOA VISTA	312	254	58	18.59%	81.41%
COMDUSA	310	254	56	18.06%	81.94%
DE LOURDES	305	258	47	15.41%	84.59%
ARIOVALDO FAVALESSA	282	220	62	21.99%	78.01%
ANTÔNIO HONÓRIO	271	221	50	18.45%	81.55%
FRADINHOS	258	210	48	18.60%	81.40%
ENSEADA DO SUÁ	235	183	52	22.13%	77.87%
SANTA HELENA	178	141	37	20.79%	79.21%
HORTO	175	133	42	24%	76%
UNIVERSITÁRIO	152	120	32	21.05%	78.95%
SEGURANÇA DO LAR	145	117	28	19.31%	80.69%
NAZARETH	135	106	29	21.48%	78.52%
MORADA DE CAMBURI	96	80	16	16.67%	83.33%
PONTAL DE CAMBURI	69	57	12	17.39%	82.61%
ILHA DO BOI	35	32	3	8.57%	91.43%
ILHA DO FRADE	10	8	2	20%	80%
AEROPORTO	8	7	1	12.50%	87.50%
ILHAS OCEÂNICAS DE TRINDADE	2	0	2	100%	0%
PARQUE INDUSTRIAL	1	1	0	0%	100%

Table A.5 — Weekday Appointment Counts

appointment_weekday	total_appointments	attended_appointments	missed_appointments	no_show_rate_pct	attended_rate_pct
Monday	22,715	18,025	4,690	20.65%	79.35%
Tuesday	25,640	20,488	5,152	20.09%	79.91%
Wednesday	25,867	20,774	5,093	19.69%	80.31%
Thursday	17,247	13,909	3,338	19.35%	80.65%
Friday	19,019	14,982	4,037	21.23%	78.77%
Saturday	39	30	9	23.08%	76.92%

Table A.6 — Weekly Appointment Data

week_start	appointment_days	total_appointments	attended_appointments	missed_appointments	no_show_rate_pct	attended_rate_pct
2016-04-25	1	3,235	2,602	633	19.57%	80.43%
2016-05-02	5	20,952	16,685	4,267	20.37%	79.63%
2016-05-09	6	21,722	17,075	4,647	21.39%	78.61%
2016-05-16	5	21,456	16,791	4,665	21.74%	78.26%
2016-05-23	2	7,918	6,348	1,570	19.83%	80.17%
2016-05-30	5	21,657	17,583	4,074	18.81%	81.19%
2016-06-06	3	13,587	11,124	2,463	18.13%	81.87%

Table A.7 — Daily Appointment Data

appointment_day	total_appointments	attended_appointments	missed_appointments	no_show_rate_pct	attended_rate_pct
2016-04-29	3,235	2,602	633	19.57%	80.43%
2016-05-02	4,376	3,515	861	19.68%	80.32%
2016-05-03	4,256	3,425	831	19.53%	80.47%
2016-05-04	4,168	3,195	973	23.34%	76.66%
2016-05-05	4,273	3,466	807	18.89%	81.11%
2016-05-06	3,879	3,084	795	20.49%	79.51%
2016-05-09	4,520	3,501	1,019	22.54%	77.46%
2016-05-10	4,308	3,316	992	23.03%	76.97%
2016-05-11	4,474	3,589	885	19.78%	80.22%
2016-05-12	4,394	3,557	837	19.05%	80.95%
2016-05-13	3,987	3,082	905	22.70%	77.30%
2016-05-14	39	30	9	23.08%	76.92%
2016-05-16	4,613	3,564	1,049	22.74%	77.26%
2016-05-17	4,372	3,437	935	21.39%	78.61%
2016-05-18	4,373	3,483	890	20.35%	79.65%
2016-05-19	4,270	3,378	892	20.89%	79.11%
2016-05-20	3,828	2,929	899	23.48%	76.52%
2016-05-24	4,009	3,198	811	20.23%	79.77%
2016-05-25	3,909	3,150	759	19.42%	80.58%
2016-05-30	4,514	3,626	888	19.67%	80.33%
2016-05-31	4,279	3,512	767	17.92%	82.08%
2016-06-01	4,464	3,652	812	18.19%	81.81%
2016-06-02	4,310	3,508	802	18.61%	81.39%
2016-06-03	4,090	3,285	805	19.68%	80.32%
2016-06-06	4,692	3,819	873	18.61%	81.39%
2016-06-07	4,416	3,600	816	18.48%	81.52%
2016-06-08	4,479	3,705	774	17.28%	82.72%

Table A.8 — Patient Characteristics and Medical Indicators

indicator	category	total_appointments	attended_appointments	missed_appointments	no_show_rate_pct
Gender	Female	71,840	57,246	14,594	20.31%
Gender	Male	38,687	30,962	7,725	19.97%
Scholarship	No	99,666	79,925	19,741	19.81%
Scholarship	Yes	10,861	8,283	2,578	23.74%
Hypertension	No	88,726	70,179	18,547	20.90%
Hypertension	Yes	21,801	18,029	3,772	17.30%
Diabetes	No	102,584	81,695	20,889	20.36%
Diabetes	Yes	7,943	6,513	1,430	18%
Alcoholism Indicator	No	107,167	85,525	21,642	20.19%
Alcoholism Indicator	Yes	3,360	2,683	677	20.15%
Handicap Indicator	0	108,286	86,374	21,912	20.24%
Handicap Indicator	1	2,042	1,676	366	17.92%
Handicap Indicator	2	183	146	37	20.22%
Handicap Indicator	3	13	10	3	23.08%
Handicap Indicator	4	3	2	1	33.33%

APPENDIX B — SQL QUERIES

B.1 — Cleaned Working Table Creation

SQL Query B.1: Create Cleaned Appointment Table

Source: `02_process_cleaning.sql` — Step 3D

```sql
-- 3D: Create the cleaned working table
CREATE TABLE public.appointments_clean AS
SELECT
  patient_id,
  appointment_id::BIGINT AS appointment_id,
  gender,
  scheduled_day::TIMESTAMPTZ AS scheduled_day,
  LEFT(appointment_day, 10)::DATE AS appointment_day,
  age::INTEGER AS age,
  neighbourhood,
  scholarship::SMALLINT AS scholarship,
  hipertension::SMALLINT AS hypertension,
  diabetes::SMALLINT AS diabetes,
  alcoholism::SMALLINT AS alcoholism,
  handcap::INTEGER AS handcap,
  sms_received::SMALLINT AS sms_received,
  no_show
FROM public.appointments_raw;
```

B.2 — Invalid Age Correction

SQL Query B.2: Replace Invalid Patient Age with NULL

Source: `02_process_cleaning.sql` — Step 4B

```sql
-- 4B: Replace invalid patient age with NULL
-- Reason: Age -1 is invalid, and the correct age cannot be verified from other
appointments.
-- Preserve the appointment and original raw record.
UPDATE public.appointments_clean
SET age = NULL
WHERE appointment_id = 5775010
 AND age = -1
RETURNING
  appointment_id,
  age;
```

B.3 — Scheduling Lead Time Calculation

SQL Query B.3: Create and Calculate Appointment Lead Days

Source: `02_process_cleaning.sql` — Steps 4D.2–4D.3

```sql
-- 4D.2: Create a column for appointment lead times
ALTER TABLE public.appointments_clean
ADD COLUMN IF NOT EXISTS lead_days INTEGER;
-- 4D.3: Calculate valid appointment lead times
UPDATE public.appointments_clean
SET lead_days = CASE
  WHEN appointment_day >=
    (scheduled_day AT TIME ZONE 'UTC')::DATE
  THEN appointment_day -
    (scheduled_day AT TIME ZONE 'UTC')::DATE
  ELSE NULL
END;
```

B.4 — No-Show Indicator Creation

SQL Query B.4: Create and Populate Numeric No-Show Indicator

Source: `02_process_cleaning.sql` — Steps 4F.1–4F.2

```sql
-- 4F.1: Create numeric no-show indicator
ALTER TABLE public.appointments_clean
ADD COLUMN IF NOT EXISTS no_show_flag SMALLINT;
-- 4F.2: Convert appointment outcomes into numeric values
-- Yes = 1 (Patient missed appointment)
-- No = 0 (Patient did not miss appointment)
UPDATE public.appointments_clean
SET no_show_flag = CASE
  WHEN no_show = 'Yes' THEN 1
  WHEN no_show = 'No' THEN 0
  ELSE NULL
END
WHERE no_show_flag IS NULL;
```

B.5 — Data Integrity Validation

SQL Query B.5.1: Verify Cleaned Dataset Integrity

Source: `02_process_cleaning.sql` — Step 5A

```sql
-- 5A: Verify cleaned dataset integrity
SELECT
  COUNT(*) AS total_records,
  COUNT(DISTINCT appointment_id)
    AS unique_appointments,
  COUNT(*) FILTER (
    WHERE age < 0
  ) AS negative_ages,
  COUNT(*) FILTER (
    WHERE age IS NULL
  ) AS missing_ages,
  COUNT(*) FILTER (
    WHERE lead_days IS NULL
  ) AS missing_lead_times,
  COUNT(*) FILTER (
    WHERE lead_days < 0
  ) AS negative_lead_times,
  COUNT(*) FILTER (
    WHERE no_show_flag IS DISTINCT FROM
      CASE
        WHEN no_show = 'Yes' THEN 1
        WHEN no_show = 'No' THEN 0
        ELSE NULL
      END
  ) AS incorrect_no_show_flags
FROM public.appointments_clean;
```

SQL Query B.5.2: Verify Appointment IDs Between Raw and Cleaned Tables

Source: `02_process_cleaning.sql` — Step 5B

```sql
-- 5B: Verify appointment IDs between raw and cleaned tables
SELECT
  COUNT(*) FILTER (
    WHERE r.appointment_id IS NOT NULL
      AND c.appointment_id IS NOT NULL
  ) AS matching_appointments,
  COUNT(*) FILTER (
    WHERE r.appointment_id IS NOT NULL
```

```
      AND c.appointment_id IS NULL
  ) AS missing_from_clean,
  COUNT(*) FILTER (
    WHERE r.appointment_id IS NULL
      AND c.appointment_id IS NOT NULL
  ) AS unexpected_in_clean
FROM public.appointments_raw AS r
FULL OUTER JOIN public.appointments_clean AS c
  ON r.appointment_id::BIGINT = c.appointment_id;
```

SQL Query B.5.3: Verify Original Field Values Against Cleaned Data

Source: `02_process_cleaning.sql` — Step 5C

```
-- 5C: Verify original field values against cleaned data
-- Account for intentional data type conversions and the correction of age -1 to NULL.
SELECT
  COUNT(*) AS checked_appointments,
  COUNT(*) FILTER (
    WHERE ROW(
        c.patient_id,
        c.appointment_id,
        c.gender,
        c.scheduled_day,
        c.appointment_day,
        c.age,
        c.neighbourhood,
        c.scholarship,
        c.hypertension,
        c.diabetes,
        c.alcoholism,
        c.handcap,
        c.sms_received,
        c.no_show
    ) IS DISTINCT FROM ROW(
        r.patient_id,
        r.appointment_id::BIGINT,
        r.gender,
        r.scheduled_day::TIMESTAMPTZ,
        LEFT(r.appointment_day, 10)::DATE,
        NULLIF(r.age, '-1')::INTEGER,
        r.neighbourhood,
        r.scholarship::SMALLINT,
```

```
        r.hipertension::SMALLINT,
        r.diabetes::SMALLINT,
        r.alcoholism::SMALLINT,
        r.handcap::INTEGER,
        r.sms_received::SMALLINT,
        r.no_show
      )
  ) AS unexpected_field_differences
FROM public.appointments_raw AS r
INNER JOIN public.appointments_clean AS c
  ON r.appointment_id::BIGINT = c.appointment_id;
```

B.6 — Age and Scheduling Lead Time Classifications

SQL Query B.6.1: Classify Appointments by Patient Age Group

Source: `03_analyze.sql` — Step 1D

```
-- Step 1D: Analyze No-Show Rates by Age Group
WITH age_groups AS (
  SELECT
    CASE
      WHEN age IS NULL THEN 'Unknown'
      WHEN age BETWEEN 0 AND 17 THEN '0-17'
      WHEN age BETWEEN 18 AND 29 THEN '18-29'
      WHEN age BETWEEN 30 AND 44 THEN '30-44'
      WHEN age BETWEEN 45 AND 59 THEN '45-59'
      WHEN age BETWEEN 60 AND 74 THEN '60-74'
      ELSE '75+'
    END AS age_group,
    no_show_flag
  FROM public.appointments_clean
)
SELECT
  age_group,
  COUNT(*) AS total_appointments,
  COUNT(*) FILTER (WHERE no_show_flag = 0) AS attended_appointments,
  COUNT(*) FILTER (WHERE no_show_flag = 1) AS missed_appointments,
  ROUND(
    100.0 * COUNT(*) FILTER (WHERE no_show_flag = 1)
    / NULLIF(COUNT(*), 0),
    2
  ) AS no_show_rate_pct
```

```
FROM age_groups
GROUP BY age_group
ORDER BY
  CASE age_group
    WHEN '0-17' THEN 1
    WHEN '18-29' THEN 2
    WHEN '30-44' THEN 3
    WHEN '45-59' THEN 4
    WHEN '60-74' THEN 5
    WHEN '75+' THEN 6
    ELSE 7
  END;
```

SQL Query B.6.2: Classify Appointments by Scheduling Lead Time

Source: `03_analyze.sql` — Step 1F

```
-- Step 1F: Analyze No-Show Rates by Scheduling Lead Time
WITH lead_time_groups AS (
  SELECT
    CASE
      WHEN lead_days IS NULL THEN 'Unknown'
      WHEN lead_days = 0 THEN 'Same Day'
      WHEN lead_days BETWEEN 1 AND 7 THEN '1-7 Days'
      WHEN lead_days BETWEEN 8 AND 14 THEN '8-14 Days'
      WHEN lead_days BETWEEN 15 AND 30 THEN '15-30 Days'
      ELSE '31+ Days'
    END AS lead_time_group,
    no_show_flag
  FROM public.appointments_clean
)
SELECT
  lead_time_group,
  COUNT(*) AS total_appointments,
  COUNT(*) FILTER (WHERE no_show_flag = 0) AS attended_appointments,
  COUNT(*) FILTER (WHERE no_show_flag = 1) AS missed_appointments,
  ROUND(
    100.0 * COUNT(*) FILTER (WHERE no_show_flag = 1)
    / NULLIF(COUNT(*), 0),
    2
  ) AS no_show_rate_pct,
  ROUND(
    100.0 * COUNT(*) FILTER (WHERE no_show_flag = 0)
```

```
    / NULLIF(COUNT(*), 0),
    2
  ) AS attended_rate_pct
FROM lead_time_groups
GROUP BY lead_time_group
ORDER BY
  CASE lead_time_group
    WHEN 'Same Day' THEN 1
    WHEN '1-7 Days' THEN 2
    WHEN '8-14 Days' THEN 3
    WHEN '15-30 Days' THEN 4
    WHEN '31+ Days' THEN 5
    ELSE 6
  END;
```

B.7 — SMS Reminder Analysis

SQL Query B.7.1: Compare SMS Reminder Status by Scheduling Lead Time

Source: `03_analyze.sql` — Step 1H

```
-- Step 1H: Analyze SMS Reminders by Scheduling Lead Time
WITH lead_time_groups AS (
  SELECT
    CASE
      WHEN lead_days IS NULL THEN 'Unknown'
      WHEN lead_days = 0 THEN 'Same Day'
      WHEN lead_days BETWEEN 1 AND 7 THEN '1-7 Days'
      WHEN lead_days BETWEEN 8 AND 14 THEN '8-14 Days'
      WHEN lead_days BETWEEN 15 AND 30 THEN '15-30 Days'
      ELSE '31+ Days'
    END AS lead_time_group,
    sms_received,
    no_show_flag
  FROM public.appointments_clean
)
SELECT
  lead_time_group,
  CASE
    WHEN sms_received = 1 THEN 'SMS Received'
    WHEN sms_received = 0 THEN 'No SMS Received'
    ELSE 'Unknown'
  END AS sms_status,
```

```
  COUNT(*) AS total_appointments,
  COUNT(*) FILTER (WHERE no_show_flag = 0)
    AS attended_appointments,
  COUNT(*) FILTER (WHERE no_show_flag = 1)
    AS missed_appointments,
  ROUND(
    100.0 * COUNT(*) FILTER (WHERE no_show_flag = 1)
    / NULLIF(COUNT(*), 0),
    2
  ) AS no_show_rate_pct,
  ROUND(
    100.0 * COUNT(*) FILTER (WHERE no_show_flag = 0)
    / NULLIF(COUNT(*), 0),
    2
  ) AS attended_rate_pct
FROM lead_time_groups
GROUP BY lead_time_group, sms_received
ORDER BY
  CASE lead_time_group
    WHEN 'Same Day' THEN 1
    WHEN '1-7 Days' THEN 2
    WHEN '8-14 Days' THEN 3
    WHEN '15-30 Days' THEN 4
    WHEN '31+ Days' THEN 5
    ELSE 6
  END,
  sms_received DESC NULLS LAST;
```

SQL Query B.7.2: Compare SMS No-Show Rates Excluding Same-Day Appointments

Source: `03_analyze.sql` — Step 1I

```
-- Step 1I: Compare SMS No-Show Rates
-- Excluding Same-Day Appointments
SELECT
  CASE
    WHEN sms_received = 1 THEN 'SMS Received'
    WHEN sms_received = 0 THEN 'No SMS Received'
    ELSE 'Unknown'
  END AS sms_status,
  COUNT(*) AS total_appointments,
  COUNT(*) FILTER (WHERE no_show_flag = 0)
    AS attended_appointments,
```

```
  COUNT(*) FILTER (WHERE no_show_flag = 1)
    AS missed_appointments,
  ROUND(
    100.0 * COUNT(*) FILTER (WHERE no_show_flag = 1)
    / NULLIF(COUNT(*), 0),
    2
  ) AS no_show_rate_pct,
  ROUND(
    100.0 * COUNT(*) FILTER (WHERE no_show_flag = 0)
    / NULLIF(COUNT(*), 0),
    2
  ) AS attended_rate_pct
FROM public.appointments_clean
WHERE lead_days >= 1
GROUP BY sms_received
ORDER BY sms_received DESC;
```

B.8 — Neighborhood Ranking Analysis

SQL Query B.8.1: Identify Neighborhoods with the Highest No-Show Rates

Source: `03_analyze.sql` — Step 1O

```
-- Step 1O: Analyze No-Show Rates by Neighborhood
SELECT
  neighbourhood,
  COUNT(*) AS total_appointments,
  COUNT(*) FILTER (WHERE no_show_flag = 0)
    AS attended_appointments,
  COUNT(*) FILTER (WHERE no_show_flag = 1)
    AS missed_appointments,
  ROUND(
    100.0 * COUNT(*) FILTER (WHERE no_show_flag = 1)
    / NULLIF(COUNT(*), 0),
    2
  ) AS no_show_rate_pct,
   ROUND(
    100.0 * COUNT(*) FILTER (WHERE no_show_flag = 0)
    / NULLIF(COUNT(*), 0),
    2
  ) AS attended_rate_pct
FROM public.appointments_clean
WHERE neighbourhood IS NOT NULL
```

```
GROUP BY neighbourhood
HAVING COUNT(*) >= 100
ORDER BY
  no_show_rate_pct DESC,
  total_appointments DESC
LIMIT 15;
```

SQL Query B.8.2: Identify Neighborhoods with the Most Missed Appointments

Source: `03_analyze.sql` — Step 1P

```
-- Step 1P: Identify Neighborhoods with the Most Missed Appointments
SELECT
  neighbourhood,
  COUNT(*) AS total_appointments,
  COUNT(*) FILTER (WHERE no_show_flag = 0)
    AS attended_appointments,
  COUNT(*) FILTER (WHERE no_show_flag = 1)
    AS missed_appointments,
  ROUND(
    100.0 * COUNT(*) FILTER (WHERE no_show_flag = 1)
    / NULLIF(COUNT(*), 0),
    2
  ) AS no_show_rate_pct,
  ROUND(
    100.0 * COUNT(*) FILTER (WHERE no_show_flag = 0)
    / NULLIF(COUNT(*), 0),
    2
  ) AS attended_rate_pct
FROM public.appointments_clean
WHERE neighbourhood IS NOT NULL
GROUP BY neighbourhood
HAVING COUNT(*) >= 100
ORDER BY
  missed_appointments DESC,
  total_appointments DESC
LIMIT 15;
```